# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Fractions** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** If $\frac{5}{9}$ of a number is 20, then $\frac{2}{3}$ of the number is: 1

**(a)** 20 **(b)** $2\frac{2}{3}$ **(c)** 36 **(d)** 24

**2.** Which is the smallest of $\frac{5}{8}$, $\frac{8}{13}$, $\frac{3}{5}$, $\frac{7}{12}$? 1

**(a)** $\frac{5}{8}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{8}{13}$ **(d)** $\frac{7}{12}$

**3.** A rod $4\frac{3}{5}$ m long has $\frac{7}{10}$ m cut off. The remaining length in metres is: 1

**(a)** $\frac{1}{10}$ **(b)** $5\frac{3}{10}$ **(c)** $3\frac{3}{10}$ **(d)** $3\frac{9}{10}$

**4.** To add $\frac{1}{6}$ and $\frac{1}{10}$, the common denominator used is the LCM of 6 and 10. The LCM is: 1

**(a)** 30 **(b)** 36 **(c)** 15 **(d)** 60

**5.** $\frac{4}{5} + \frac{7}{10} - \frac{3}{4} =$ 1

**(a)** $\frac{17}{20}$ **(b)** $\frac{3}{4}$ **(c)** $1\frac{3}{5}$ **(d)** $2\frac{1}{4}$

**6.** $\frac{4}{5}$ of a number minus $\frac{3}{10}$ of it is 20. The number is: 1

**(a)** 50 **(b)** 18 **(c)** 20 **(d)** 40

**7.** A drum is $\frac{4}{5}$ full. 21 litres are removed and it becomes $\frac{3}{10}$ full. The capacity (litres) is: 1

**(a)** 42 **(b)** 43 **(c)** 41 **(d)** 52

**8.** A rod 10 m long is cut into pieces of $\frac{5}{6}$ m each. The number of pieces is: 1

**(a)** 11 **(b)** 12 **(c)** 14 **(d)** 13

**9.** **Assertion (A):** $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$. 1

**Reason (R):** Unlike fractions are first changed into equivalent fractions with a common denominator.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** $\frac{2}{3}$ of 12 is 8. 1
**Reason (R):** Every proper fraction is less than 1.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A boy walks $\frac{7}{8}$ km to school and $\frac{3}{10}$ km to a shop, then returns by the same route. Find the total distance. 2

**12.** $\frac{4}{9}$ of a number is 16. Find the number and its $\frac{3}{4}$. 2

**13.** Find the fraction which must be added to $\frac{4}{15}$ to get $\frac{9}{10}$. 2

**14.** A boy walks $\frac{2}{3}$ km to school and $\frac{5}{9}$ km to a shop, then returns by the same route. Find the total distance. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A school has 540 students. $\frac{5}{6}$ of them are girls. $\frac{2}{9}$ of the girls play chess. How many girls play chess and how many boys are there? 3

**16.** Find the missing numerator ?: $\frac{?}{18} + \frac{1}{9} = \frac{5}{6}$. 3

**17.** Pipe A fills $\frac{2}{9}$ of a cistern in one hour and pipe B fills $\frac{1}{3}$ of it in one hour. What part of the cistern is filled in one hour when both work together? What part remains empty? 3

**OR**

A school has 300 students. $\frac{3}{5}$ of them are girls. $\frac{5}{6}$ of the girls play chess. How many girls play chess and how many boys are there? 3

**18.** Find the missing numerator ?: $\frac{?}{16} + \frac{1}{4} = \frac{7}{8}$. 3

**OR**

Pipe A fills $\frac{1}{7}$ of a cistern in one hour and pipe B fills $\frac{1}{4}$ of it in one hour. What part of the cistern is filled in one hour when both work together? What part remains empty? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** There are 40 students in a class. $\frac{3}{5}$ of them like cricket, $\frac{1}{4}$ of them like football, and the rest like other games. 5
(i) How many students like cricket?
(ii) How many like football?
(iii) How many like other games?
(iv) What fraction of the class likes other games? Show it in simplest form.

**20.** A trader sold $\frac{1}{5}$ of his cloth on day 1 and $\frac{3}{8}$ of the remaining on day 2. If 25 metres are left, find the total length of cloth. 5

**OR**

Add $4\frac{1}{3}$, $2\frac{1}{4}$ and $1\frac{1}{6}$ and subtract $\frac{11}{12}$ from the sum. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**